Truth Lemma for the term model (no identity)

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the statement's FOL branch restricts to sentences without identity; the identity-language version is the sibling lem:truth at source line 1876 (quotient term model), collected separately

Editorial note. The label lem:truth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Term model and term values. Same unit, \ollabel{defn:termmodel} (source line 1186) and \ollabel{lem:val-in-termmodel} (registered card). Supplies M(Gamma*) and the fact that every closed term denotes itself, which the atomic and quantifier cases read off.
- Closure clauses for complete consistent sets. Same unit, \ollabel{prop:ccs} (registered card). Converts the membership facts assembled by the induction hypothesis into membership of the compound sentence in the connective cases.
- Saturation instances. Same unit, \ollabel{prop:saturated-instances} (source line 1648). Closes the quantifier cases: all instances in Gamma* iff the universal sentence belongs; an instance in Gamma* iff the existential belongs.

Source reference. Original proposition prop-quant-termmodel: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 1265–1265. Prerequisite pointer invoked at proof line(s) 1392; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 822–822. Prerequisite pointer invoked at proof line(s) 1360; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-and: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 828–828. Prerequisite pointer invoked at proof line(s) 1360; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-if: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 834–834. Prerequisite pointer invoked at proof line(s) 1384; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ccs-or: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 831–831. Prerequisite pointer invoked at proof line(s) 1373; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-saturated-instances: sources/olp-first-order-logic-completeness/pinned-original-source.tex, source lines 1648–1648. Prerequisite pointer invoked at proof line(s) 1394; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Truth Lemma)**.** *Suppose A does not contain $=$. Then$\mathfrak{M}(\Gamma^*) \vDash$ iff $A \in \Gamma^*$.*

Proof. We prove both directions simultaneously, and by induction on A.

1. $A \equiv \bot$: $\mathfrak{M}(\Gamma^*) \nvDash \bot$ by definition of satisfaction. On the other hand, $\bot \notin \Gamma^*$ since Γ^* is consistent.

2. $A \equiv \top$: $\mathfrak{M}(\Gamma^*) \vDash$ by definition of satisfaction. On the other hand, $\top \in \Gamma^*$ since Γ^* is consistent and complete, and $\Gamma^* \vdash \top$.

3. $A \equiv R(t_1, \ldots, t_n)$: $\mathfrak{M}(\Gamma^*) \vDash R(t_1, \ldots, t_n)$ iff $\langle t_1, \ldots, t_n \rangle \in sRM(\Gamma^*)$ (by the definition of satisfaction) iff $R(t_1, \ldots, t_n) \in \Gamma^*$ (by the construction of $\mathfrak{M}(\Gamma^*)$).

4. $A \equiv \neg B$: $\mathfrak{M}(\Gamma^*) \vDash$ iff $\mathfrak{M}(\Gamma^*) \nvDash B$ (by definition of satisfaction). By induction hypothesis, $\mathfrak{M}(\Gamma^*) \nvDash B$ iff $B \notin \Gamma^*$. Since Γ^* is consistent and complete, $B \notin \Gamma^*$ iff $\neg B \in \Gamma^*$.

5. $A \equiv B \wedge C$: exercise.

6. $A \equiv B \vee C$: exercise.

7. $A \equiv B \to C$: exercise.

8. $A \equiv \forall x\, B(x)$: exercise.

9. $A \equiv \exists x\, B(x)$: exercise.

□